Ice cream gets its different flavors from both **natural and synthetic compounds.**

Compounds

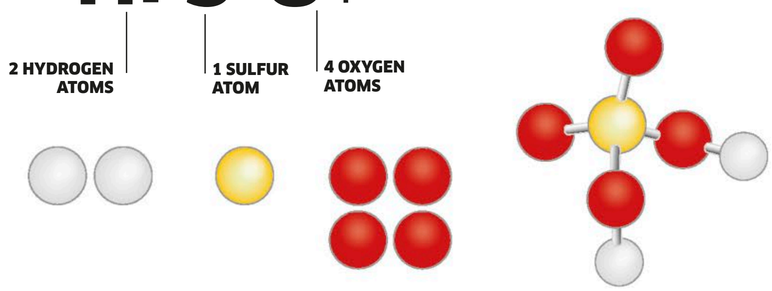
When two or more elements join together by forming chemical bonds, they make up a new, different substance. This substance is known as a compound.

Compounds are not just mixtures of elements. A mixture can be separated into the individual substances it contains, but it is not easy to turn a compound back into the elements that formed it. For example, water is a compound of hydrogen and oxygen. Only through a chemical reaction can it be changed back into these separate elements. A compound is made of atoms of two or more elements in a particular ratio. In water, for example, the ratio is two hydrogen atoms and one oxygen atom for every water molecule.

Fascinating formula

The chemical formula of a compound tells you which elements are present, and in what ratio. The compound sulfuric acid (H_2SO_4) is made of molecules that each contain two hydrogen atoms, one sulfur atom, and four oxygen atoms.

$$H_2SO_4 = \text{Sulfuric acid}$$



Great ways to bond

There are two types of bond that can hold the atoms in a compound together: covalent and ionic (see pp.16–17). Covalent bonds form between non-metal atoms. Ionic bonds form between metal and non-metal atoms.

Covalent compounds
Covalent compounds, such as sugar, form molecules in which the atoms form covalent bonds. They melt and boil at lower temperatures than ionic compounds. When they dissolve in water, they do not conduct electricity.

Salt lowers the freezing point of water, so it is used for melting ice and snow on roads.

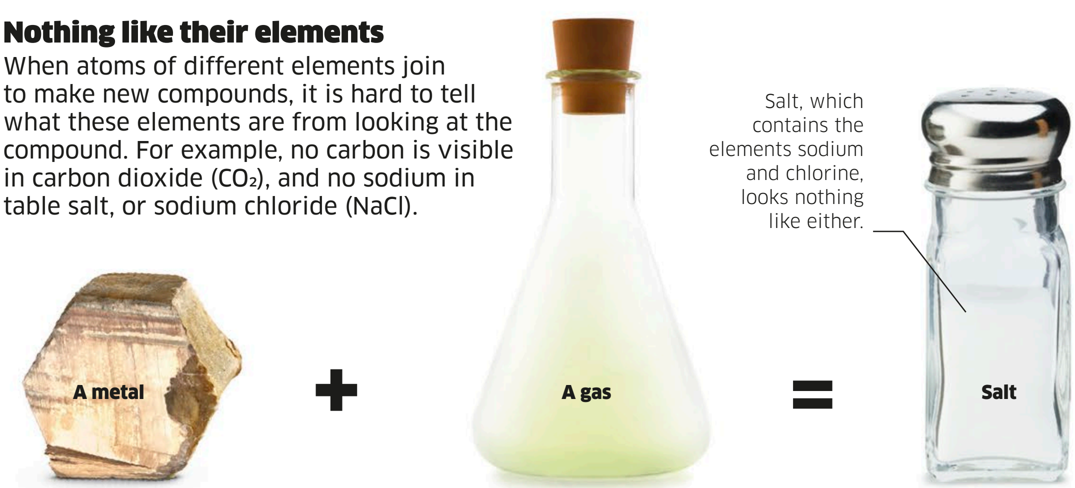
Ionic compounds
Ionic compounds consist of ions. An ion is an electrically charged particle, formed when an atom has lost or gained electrons. Ions bond together, forming crystals with high melting points. Salt is an ionic compound.

Calcium carbonate is found in egg shells, but also in harder seashells.

Best of both
Most compounds combine ionic and covalent bonding. In calcium carbonate, for example, calcium ions form ionic bonds with carbonate ions. Each carbonate ion contains carbon and oxygen atoms held together by covalent bonds.

Nothing like their elements

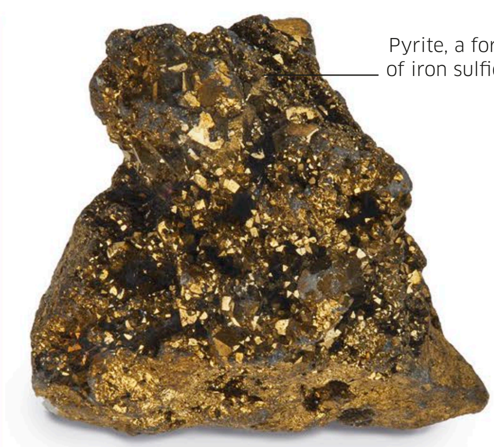
When atoms of different elements join to make new compounds, it is hard to tell what these elements are from looking at the compound. For example, no carbon is visible in carbon dioxide (CO_2), and no sodium in table salt, or sodium chloride (NaCl).



Na Sodium + **Cl** Chlorine = **NaCl** Sodium chloride

Look what they have become
In chemical reactions, atoms from different elements regroup into new, different atom combinations. The resulting substances often look, and feel, completely different, too. For instance, sodium is a shiny metal, and chlorine is a pale green gas, but together they make sodium chloride (salt), a white crystal.

Pyrite, a form of iron sulfide



Iron sulfide
Iron sulfide, a compound of iron and sulfur, exists in several forms. Iron filings and yellow sulfur powder can be fused together to form a black solid called iron (II) sulfide (FeS). The mineral pyrite (FeS_2, above), known as “fool’s gold,” is another form of iron sulfide. Unlike iron, neither of these compounds is magnetic.